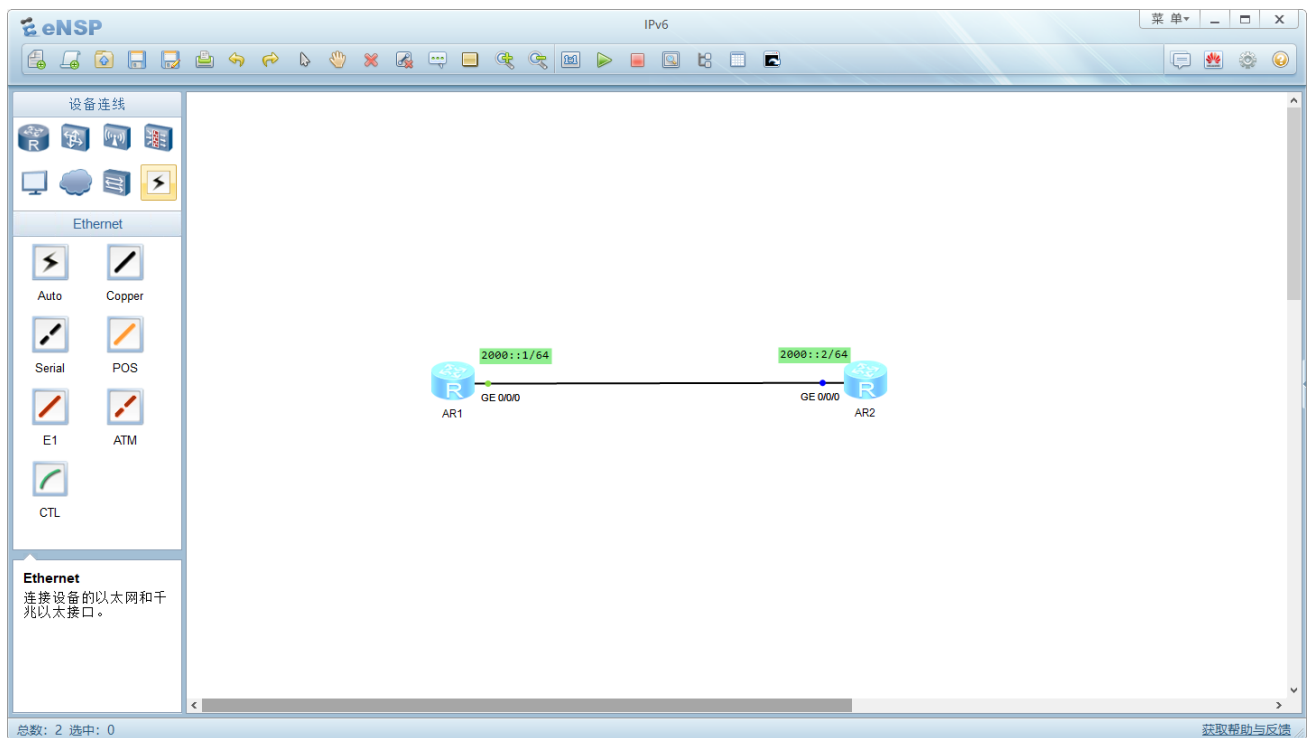
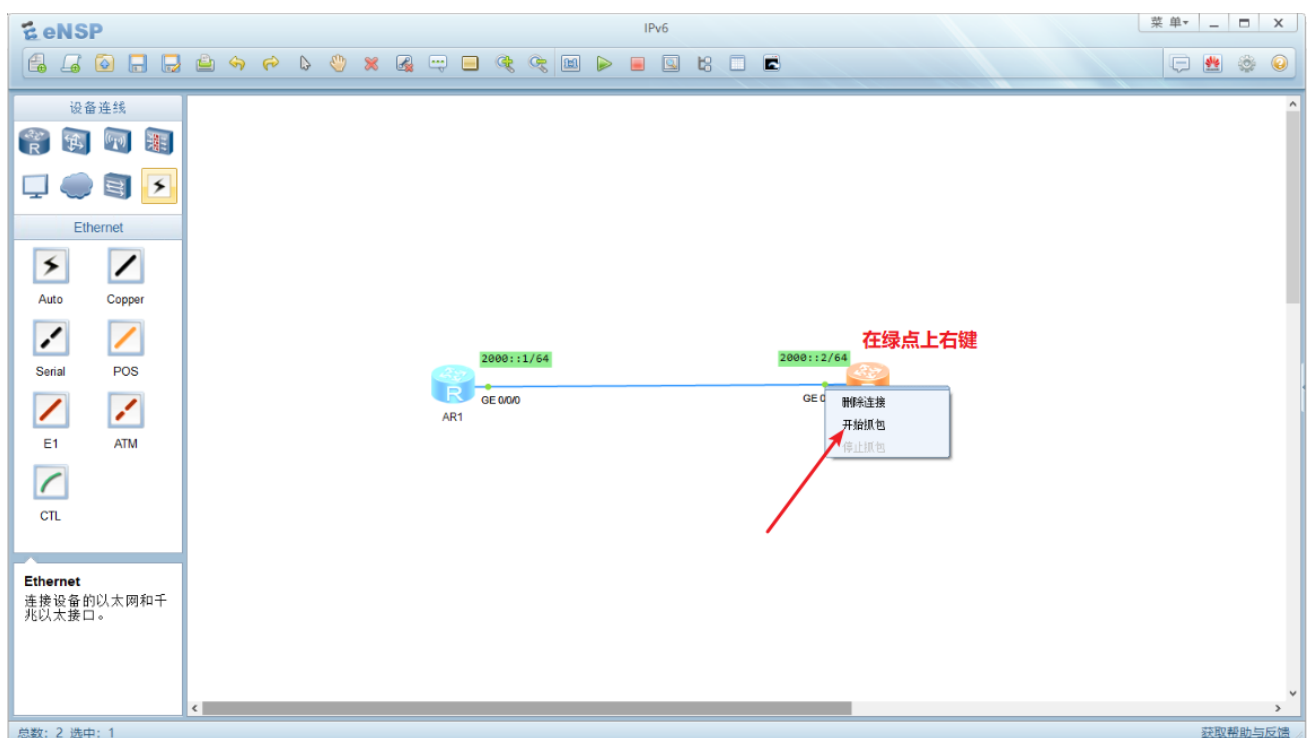


一、实验准备

1、根据平台提供的地址，登录操作机。使用 ensp 打开“IPv6 实验拓扑”文件夹下的拓扑图。



2、在 AR2 的 G0/0/0 接口上打开 wireshark 抓包。



3、在 AR1 上 ping AR2 的地址，获得 IPv6 的数据包。

The screenshot shows a network simulator interface. The top window is a terminal for AR1, and the bottom window is a packet capture tool.

Terminal Window (AR1):

```
The current configuration will be written to the device.
Are you sure to continue? (y/n)[n]:y
It will take several minutes to save configuration file, please wait.....
Configuration file had been saved successfully
Note: The configuration file will take effect after being activated
<AR1>
<AR1>
<AR1>
<AR1>
<AR1>
<AR1>ping ipv6 2000::2
PING 2000::2 : 56 data bytes, press CTRL_C to break
  Reply from 2000::2
    bytes=56 Sequence=1 hop limit=64  time = 40 ms
  Reply from 2000::2
    bytes=56 Sequence=2 hop limit=64  time = 10 ms
  Reply from 2000::2
    bytes=56 Sequence=3 hop limit=64  time = 30 ms
  Reply from 2000::2
    bytes=56 Sequence=4 hop limit=64  time = 30 ms
  Reply from 2000::2
    bytes=56 Sequence=5 hop limit=64  time = 20 ms

--- 2000::2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 10/26/40 ms

<AR1>
```

Packet Capture Window:

No.	Time	Source	Destination	Protocol	Length	Data
1	1970-01-04 07:12:35.093000	2000::1	2000::2	ICMPv6	118	Echo (ping) request id=0xd2ab, seq=256, hop limit=64 (request in 2)
2	1970-01-04 07:12:35.093000	2000::1	2000::2	ICMPv6	118	Echo (ping) reply id=0xd2ab, seq=256, hop limit=64 (reply in 1)
3	1970-01-04 07:12:35.093000	2000::1	2000::2	ICMPv6	118	Echo (ping) request id=0xd2ab, seq=512, hop limit=64 (request in 4)
4	1970-01-04 07:12:35.578000	2000::2	2000::1	ICMPv6	118	Echo (ping) reply id=0xd2ab, seq=512, hop limit=64 (reply in 3)
5	1970-01-04 07:12:36.078000	2000::1	2000::2	ICMPv6	118	Echo (ping) request id=0xd2ab, seq=768, hop limit=64 (request in 6)
6	1970-01-04 07:12:36.078000	2000::2	2000::1	ICMPv6	118	Echo (ping) reply id=0xd2ab, seq=768, hop limit=64 (reply in 5)
7	1970-01-04 07:12:36.562000	2000::1	2000::2	ICMPv6	118	Echo (ping) request id=0xd2ab, seq=1024, hop limit=64 (request in 8)
8	1970-01-04 07:12:36.578000	2000::2	2000::1	ICMPv6	118	Echo (ping) reply id=0xd2ab, seq=1024, hop limit=64 (reply in 7)
9	1970-01-04 07:12:37.042000	2000::1	2000::2	ICMPv6	118	Echo (ping) request id=0xd2ab, seq=1280, hop limit=64 (request in 10)
10	1970-01-04 07:12:37.062000	2000::2	2000::1	ICMPv6	118	Echo (ping) reply id=0xd2ab, seq=1280, hop limit=64 (reply in 9)
11	1970-01-04 07:12:40.125000	fe80::2e0:fcff:feb2:6cd6	2000::1	ICMPv6	86	Neighbor Solicitation for 2000::1 from 00:e0:fc:b2:6c:d6
12	1970-01-04 07:12:40.125000	2000::1	fe80::2e0:fcff:feb2:6cd6	ICMPv6	86	Neighbor Advertisement 2000::1 (rtr, sol, ovr) is at 00:e0:fc:b2:6c:d6
13	1970-01-04 07:12:41.031000	fe80::2e0:fcff:feb2:35b0	2000::2	ICMPv6	86	Neighbor Solicitation for 2000::2 from 00:e0:fc:b2:6c:d6
14	1970-01-04 07:12:41.031000	2000::2	fe80::2e0:fcff:feb2:35b0	ICMPv6	86	Neighbor Advertisement 2000::2 (rtr, sol, ovr) is at 00:e0:fc:b2:6c:d6
15	1970-01-04 07:12:46.046000	fe80::2e0:fcff:feb2:35b0	fe80::2e0:fcff:feb2:6cd6	ICMPv6	86	Neighbor Solicitation for fe80::2e0:fcff:feb2:6cd6 from 00:e0:fc:b2:6c:d6
16	1970-01-04 07:12:46.046000	fe80::2e0:fcff:feb2:6cd6	fe80::2e0:fcff:feb2:35b0	ICMPv6	86	Neighbor Advertisement fe80::2e0:fcff:feb2:6cd6 (rtr, sol, ovr) is at 00:e0:fc:b2:6c:d6
17	1970-01-04 07:12:46.125000	fe80::2e0:fcff:feb2:6cd6	fe80::2e0:fcff:feb2:35b0	ICMPv6	86	Neighbor Solicitation for fe80::2e0:fcff:feb2:35b0 from 00:e0:fc:b2:6c:d6
18	1970-01-04 07:12:46.125000	fe80::2e0:fcff:feb2:35b0	fe80::2e0:fcff:feb2:6cd6	ICMPv6	86	Neighbor Advertisement fe80::2e0:fcff:feb2:35b0 (rtr, sol, ovr) is at 00:e0:fc:b2:6c:d6

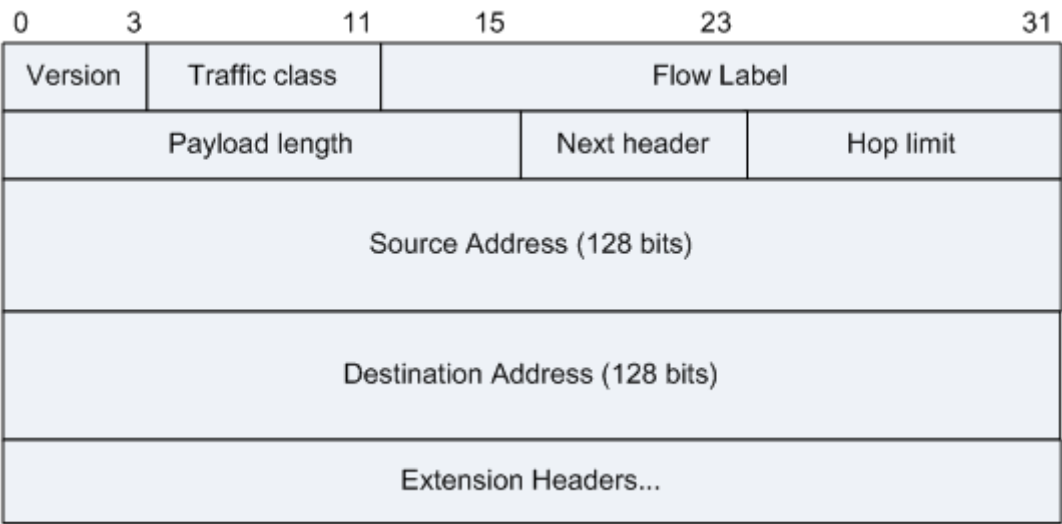
Packet Details:

```
> Frame 1: 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface 0
> Ethernet II, Src: HuaweiTe_26:35:b0 (00:e0:fc:b2:35:b0), Dst: HuaweiTe_b2:6c:d6 (00:e0:fc:b2:6c:d6)
> Internet Protocol Version 6, Src: 2000::1, Dst: 2000::2
> Internet Control Message Protocol v6
```

4、也可以直接打开“IPv6 实验拓扑”文件夹下的“IPv6.pcap”文件，直接分析。

二、实验步骤

1. IPv6 分基本头部和拓展头部。相较于 IPv4，IPv6 去掉了一些头部，把这些头部全部弄到了后面的扩展头部中。其中 IPv6 报文头格式如下。



其中各字段的含义如下：

字段	长度	含义
Version	4 比特	4：表示为 IPV4 6：表示为 IPV6
Traffic class	8 比特	流量类别。该字段及其功能类似于 IPv4 的业务类型字段。该字段以区分业务编码点（DSCP）标记一个 IPv6 数据包，以此指明数据包应当如何处理。
Flow Label	20 比特	流标签。该字段用来标记 IP 数据包的一个流，当前的标准中没有定义如何管理和处理流标签的细节。
Payload length	16 比特	该字段表示有效载荷的长度，有效载荷是指紧跟 IPv6 基本报头的数据包，包含 IPv6 扩展报头。
Next header	8 比特	下一报头，该字段指明了跟随在 IPv6 基本报头后的扩展报头的信息类型。
Hop limit	8 比特	跳数限制，该字段定义了 IPv6 数据包所能经过的最大跳数，这个字段和 IPv4 中的 TTL 字段非常相似。
Source Address	128 比特	该字段表示该报文的源地址。
Destination Address	128 比特	该字段表示该报文的目的地地址。

Extension Headers	可变	扩展报头。IPv6 取消了 IPv4 报头中的选项字段，并引入了多种扩展报文头，在提高处理效率的同时还增强了 IPv6 的灵活性，为 IP 协议提供了良好的扩展能力。当超过一种扩展报头被用在同一个分组里时，报头必须按照下列顺序出现： • IPv6 基本报头 • 逐跳选项扩展报头 • 目的选项扩展报头 • 路由扩展报头 • 分片扩展报头 • 授权扩展报头 • 封装安全有效载荷扩展报头 • 目的选项扩展报头（指那些将被分组报文的最终目的地处理的选项。） • 上层扩展报头 不是所有的扩展报头都需要被转发路由设备查看和处理的。路由设备转发时根据基本报头中 Next Header 值来决定是否要处理扩展头。 除了目的选项扩展报头出现两次（一次在路由扩展报头之前，另一次在上层扩展报头之前），其余扩展报头只出现一次。
--------------------------	-----------	---

2. 对实验准备中，抓取到的 IPv6 数据包进行分析。双击第一个数据包，打开数据包详情。

IPv6.pcap

文件(F) 编辑(E) 视图(V) 捕获(C) 分析(A) 统计(S) 电话(Y) 无线(W) 工具(T) 帮助(H)

应用显示过滤器: <Ctrl-F>

No.	Time	Source	Destination	Protocol	Length	Info
1	1970-01-04 07:57:03.718000	2000::1	2000::2	ICMPv6	118	Echo (ping) request id=0xd2ab, seq=256, hop limit=64
2	1970-01-04 07:57:03.718000	2000::2	2000::1	ICMPv6	118	Echo (ping) reply id=0xd2ab, seq=256, hop limit=64
3	1970-01-04 07:57:04.218000	2000::1	2000::2	ICMPv6	118	Echo (ping) request id=0xd2ab, seq=512, hop limit=64
4	1970-01-04 07:57:04.218000	2000::2	2000::1	ICMPv6	118	Echo (ping) reply id=0xd2ab, seq=512, hop limit=64
5	1970-01-04 07:57:04.718000	2000::1	2000::2	ICMPv6	118	Echo (ping) request id=0xd2ab, seq=768, hop limit=64
6	1970-01-04 07:57:04.718000	2000::2	2000::1	ICMPv6	118	Echo (ping) reply id=0xd2ab, seq=768, hop limit=64
7	1970-01-04 07:57:05.203000	2000::1	2000::2	ICMPv6	118	Echo (ping) request id=0xd2ab, seq=1024, hop limit=64
8	1970-01-04 07:57:05.218000	2000::2	2000::1	ICMPv6	118	Echo (ping) reply id=0xd2ab, seq=1024, hop limit=64
9	1970-01-04 07:57:05.703000	2000::1	2000::2	ICMPv6	118	Echo (ping) request id=0xd2ab, seq=1280, hop limit=64
10	1970-01-04 07:57:05.703000	2000::2	2000::1	ICMPv6	118	Echo (ping) reply id=0xd2ab, seq=1280, hop limit=64
11	1970-01-04 07:57:09.265000	fe80::2e0:fcff:feb2:6cd6	2000::1	ICMPv6	86	Neighbor Solicitation for 2000::1 from 00:e0:fc:b2:6c:d6
12	1970-01-04 07:57:09.281000	2000::1	fe80::2e0:fcff:feb2:6cd6	ICMPv6	86	Neighbor Advertisement 2000::1 (rtr, sol, ovr)
13	1970-01-04 07:57:15.265000	fe80::2e0:fcff:feb2:6cd6	fe80::2e0:fcff:feb2:6cd6	ICMPv6	86	Neighbor Solicitation for fe80::2e0:fcff:feb2:6cd6
14	1970-01-04 07:57:15.281000	fe80::2e0:fcff:feb2:6cd6	fe80::2e0:fcff:feb2:6cd6	ICMPv6	86	Neighbor Advertisement fe80::2e0:fcff:feb2:6cd6
15	1970-01-04 07:57:21.281000	fe80::2e0:fcff:feb2:6cd6	fe80::2e0:fcff:feb2:6cd6	ICMPv6	86	Neighbor Solicitation for fe80::2e0:fcff:feb2:6cd6
16	1970-01-04 07:57:21.296000	fe80::2e0:fcff:feb2:6cd6	fe80::2e0:fcff:feb2:6cd6	ICMPv6	86	Neighbor Advertisement fe80::2e0:fcff:feb2:6cd6

< Frame 1: 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface 0
> Ethernet II, Src: HuaweiTe_26:35:b0 (00:e0:fc:b2:6c:d6), Dst: HuaweiTe_b2:6c:d6 (00:e0:fc:b2:6c:d6)
> Internet Protocol Version 6, Src: 2000::1, Dst: 2000::2
> Internet Control Message Protocol v6

IPv6.pcap 分组: 16 · 已显示: 16 (100.0%) 配置: Default

Wireshark · 分组 1 · IPv6.pcap

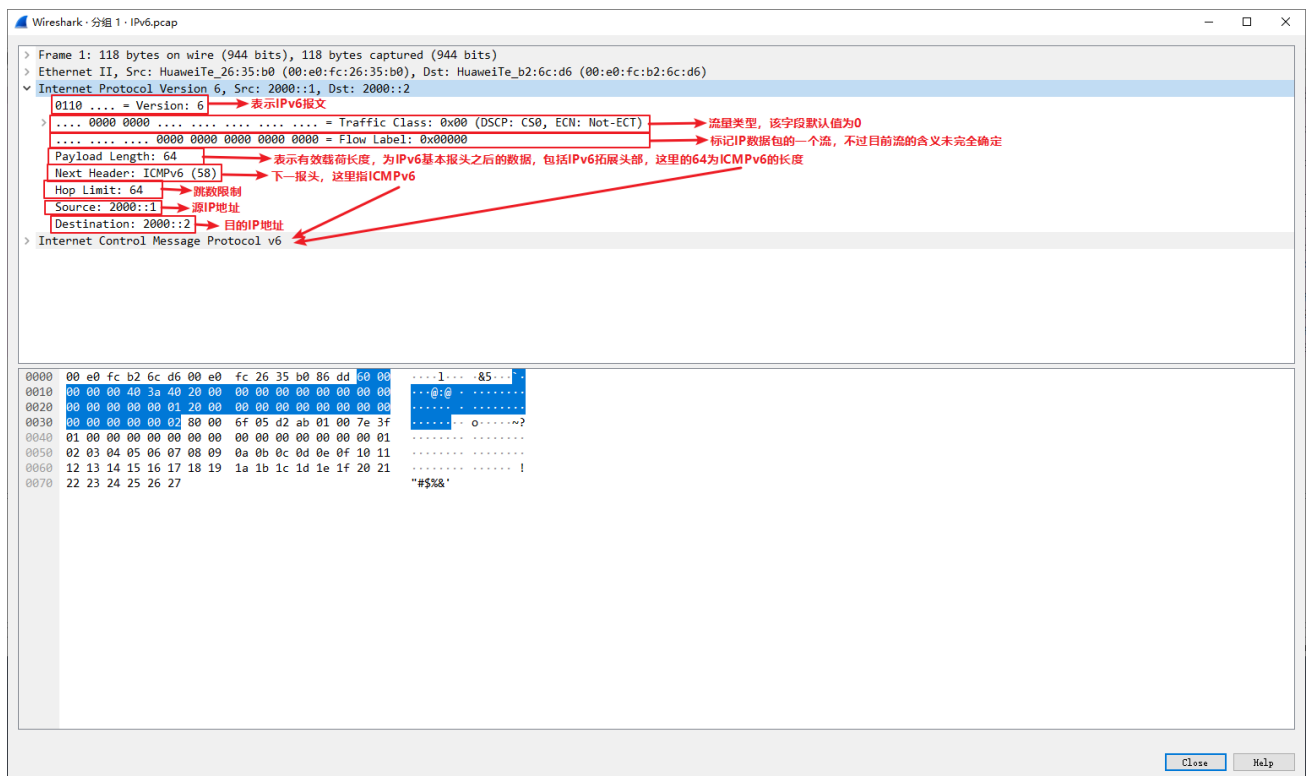
> Frame 1: 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface 0
> Ethernet II, Src: HuaweiTe_26:35:b0 (00:e0:fc:b2:6c:d6), Dst: HuaweiTe_b2:6c:d6 (00:e0:fc:b2:6c:d6)
> Internet Protocol Version 6, Src: 2000::1, Dst: 2000::2
 0110 = Version: 6
 > 0000 0000 = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
 0000 0000 0000 0000 0000 0000 = Flow Label: 0x000000
 Payload Length: 64
 Next Header: ICMPv6 (58)
 Hop Limit: 64
 Source: 2000::1
 Destination: 2000::2
> Internet Control Message Protocol v6

0000 00 e0 fc b2 6c d6 00 e0 fc 26 35 b0 86 dd 60 001...85...
0010 00 00 00 40 3a 40 20 00 00 00 00 00 00 00 00 ..@:.....
0020 00 00 00 00 00 01 20 00 00 00 00 00 00 00 00
0030 00 00 00 00 00 02 80 00 6f 05 d2 ab 01 00 7e 3fo.....?
0040 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01
0050 02 03 04 05 06 07 08 09 0a 0b 0c 0d 0e 0f 10 11
0060 12 13 14 15 16 17 18 19 1a 1b 1c 1d 1e 1f 20 21!
0070 22 23 24 25 26 27 "#\$%&'

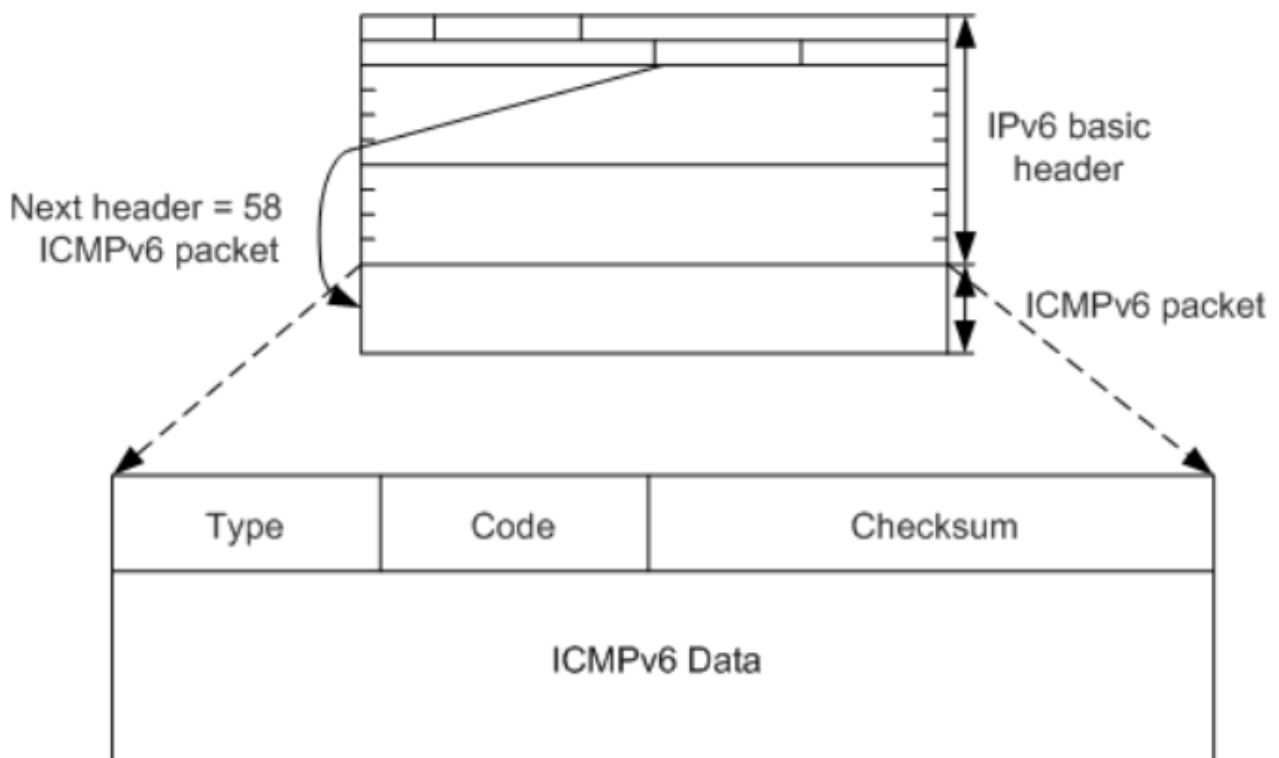
No. 1 · Time: 1970-01-04 07:57:03.718000 · Source: 2000::1 · Destination: 2000::2 · Protocol: ICMPv6 · Length: 118 · Info: Echo (ping) request id=0xd2ab, seq=256, hop limit=64 (reply in 2)

Close Help

3. 对数据包的网络层的 IPv6 协议的字段内容进行分析



4. ICMPv6 (Internet Control Message Protocol for the IPv6) 是 IPv6 的基础协议之一。在 IPv6 中, ICMPv6 除了提供 ICMPv4 常用的功能之外, 还是其它一些功能的基础, 如邻接点发现、无状态地址配置 (包括重复地址检测)、PMTU 发现等。ICMPv6 的报文格式如下:



报文中字段解释如下：

Type：表明消息的类型，0 至 127 表示差错报文类型，128 至 255 表示消息报文类型。

Code：表示此消息类型细分的类型。

Checksum：表示 ICMPv6 报文的校验和。

5. 分析步骤 3 中的 ICMPv6 报文

Wireshark · 分组 1 · IPv6.pcap

> Frame 1: 118 bytes on wire (944 bits), 118 bytes captured (944 bits)

> Ethernet II, Src: HuaweiTe_26:35:b0 (00:e0:fc:26:35:b0), Dst: HuaweiTe_b2:6c:d6 (00:e0:fc:b2:6c:d6)

> Internet Protocol Version 6, Src: 2000::1, Dst: 2000::2

> Internet Control Message Protocol v6

Type: Echo (ping) request (128) → 报文类别分为两类，1-127位错误报文 128-255位消息报文，其中128为回送请求报文

Code: 0 → 表示此类消息的细分类别

Checksum: 0x6f05 [correct] → 表示ICMPv6报文的校验和

[Checksum Status: Good]

Identifier: 0xd2ab

Sequence: 256

[Response In: 2]

> Data (56 bytes)

0000 00 e0 fc b2 6c d6 00 e0 fc 26 35 b0 86 dd 60 001...&5...
0010 00 00 00 40 3a 40 20 00 00 00 00 00 00 00 00 ...@: @
0020 00 00 00 00 00 01 20 00 00 00 00 00 00 00 00
0030 00 00 00 00 00 02 80 00 6f 05 d2 ab 01 00 7e 3fo.....?
0040 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01
0050 02 03 04 05 06 07 08 09 0a 0b 0c 0d 0e 0f 10 11!
0060 12 13 14 15 16 17 18 19 1a 1b 1c 1d 1e 1f 20 21!
0070 22 23 24 25 26 27#5%&'

Close Help

6. 实验结束